

Ontogenetic Salinity Preference Modifier

Mathematical Formulation and Rationale

Model Documentation

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1 Introduction

In individual-based models of fish movement, routing decisions are primarily driven by node suitability, classically defined by optimal growth and minimum predation mortality. However, as anadromous or estuarine species grow, they exhibit ontogenetic shifts in salinity tolerance and preference.

The Salinity Preference Modifier introduces an environmental bias to baseline node suitability. This modifier acts as a scalar value (centered at 1.0) applied to the base node fitness. It provides a biological “nudge”—encouraging out-migration of larger fish toward marine environments and retention of smaller fish in freshwater—without strictly overriding primary growth and survival drivers.

2 Mathematical Formulation

The algorithm employs a **Directional Pull Model**. Rather than treating fish preference as a specific target salinity to be matched (which erroneously penalizes purely fresh or purely marine water for transitional fish), it projects both the biological preference and the environmental state onto a bipolar scale $[-1, 1]$, where -1 represents purely freshwater and $+1$ represents purely marine water.

2.1 Directional Fish Preference

The ontogenetic shift is modeled using a modified logistic sigmoid function parameterized by the fish’s fork length (L). The baseline sigmoid is shifted and scaled such that its codomain is $[-1, 1]$.

$$P(L) = \left(\frac{2}{1 + e^{-k(L-L_{50})}} \right) - 1 \quad (1)$$

Where:

- $P(L) \in [-1, 1]$ is the directional preference.
- L is the individual’s current fork length (mm).
- L_{50} is the attraction length threshold (the length at which preference shifts from fresh to salt).
- k is the sigmoid steepness, controlling how abruptly the preference changes around L_{50} .

2.2 Directional Environmental Salinity

The target node's salinity is normalized linearly around a calculated midpoint. The model establishes a maximum functional salinity ($Sal_{max} = 32.0$ ppt) and a functional midpoint ($Sal_{mid} = 16.0$ ppt).

$$S_{env} = \frac{\min(Sal, Sal_{max}) - Sal_{mid}}{Sal_{mid}} \quad (2)$$

Where:

- $S_{env} \in [-1, 1]$ is the directional salinity of the node.
- Sal is the raw salinity at the node in parts per thousand (ppt).

2.3 Preference-Environment Match

The directional match M between the fish's biological preference and the node's environmental state is the product of the two directional variables:

$$M = P(L) \times S_{env} \quad (3)$$

Because both inputs are constrained to $[-1, 1]$:

- **Positive Match** ($M > 0$): Alignment. (e.g., small fish in fresh water, or large fish in salt water).
- **Negative Match** ($M < 0$): Conflict. (e.g., small fish in salt water, or large fish in fresh water).
- **Neutral** ($M = 0$): Occurs if either the fish is at exactly L_{50} or the water is exactly at Sal_{mid} .

2.4 Scaled Bounded Multiplier (Bias)

To prevent the salinity response from generating negative suitabilities or entirely overriding dominant growth/mortality factors, the match M is mapped to a strictly positive scalar using a global bias weight ($W \in [0, 1]$).

$$Bias = 1 + W \times M \quad (4)$$

The resulting *Bias* acts as a multiplier bounded between $1 - W$ and $1 + W$. For example, a $W = 0.2$ strictly bounds the maximum suitability modification between a 20% penalty (0.8) and a 20% bonus (1.2).

3 Implementation Rationale: Directional Pull vs. Target-Matching

Initial designs for this subsystem explored a *Target-Matching* approach, computing an absolute difference between a normalized $[0, 1]$ preference and a $[0, 1]$ environmental state.

Target-Matching proved biologically inappropriate for this use case. Under Target-Matching, a fish slightly larger than L_{50} (yielding a weak salt preference of 0.6) evaluating a weakly saline node (e.g., 13 ppt, normalized to ~ 0.4) computes a strong match due to proximity in numerical space. It establishes an artificial "ideal" salinity for every size class.

The **Directional Pull** model resolves this artifact. A 70mm fish evaluating a 13 ppt node will yield a weakly positive preference multiplied by a weakly negative environmental

state, properly resulting in a slight penalty. This ensures the macro-behavior of the system uniformly pulls large fish toward the ocean and small fish toward riverine origins.

4 Model Parameters

Parameter Key	Symbol	Description	Typical Range
SalinityAttractionLength	L_{50}	Length of neutral salinity preference (mm)	60.0 – 75.0
SalinitySigmoidSteepness	k	Rate of ontogenetic transition	0.05 – 0.15
SalinityBiasWeight	W	Maximum scaling effect on base suitability	0.10 – 0.30
SALINITY_MAX	Sal_{max}	Upper bound for salinity normalization (ppt)	32.0

Table 1: Parameters governing the salinity preference response modifier.